

STA447/2006: Stochastic Processes

Lecture Notes

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Abstract

This is a set of lecture notes prepared for the course STA447/2006 at the University of Toronto and was taught by Professor Wenlong Mou. These are roughly based on Wenlong Mou's lecture material. The references for this course are [Ros06], [Law95], [Dur12].

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Chapter 1

Markov Chains (Discrete Space)

1.1 Markov definitions and transition probabilities

1.1.1 Discrete-Time Markov Chains

Definition 1.1.1 (Discrete-time Markov chain). A discrete-time, discrete-space, time-homogeneous Markov chain is specified by the following:

- (i) A state space S , where S is assumed to be either finite or countably infinite
- (ii) Initial distribution $\{\nu_i\}_{i \in S}$, where ν_i is the probability of starting at i (at time 0). Note that: $\nu_i \geq 0$ and $\sum_i \nu_i = 1$
- (iii) Transition probabilities $\{p_{ij}\}_{i,j \in S}$ ($S \times S$ matrix), where p_{ij} is the probability of going to state j if you start at state i . Note that $p_{ij} \geq 0$ and $\sum_j p_{ij} = 1$

For a Markov chain, X_n is the Markov chain's state at time $t = n$. So, naturally $X_0, X_1, \dots$ are random variables as well. At time $t = 0$, we have $\mathbb{P}(X_0 = i) = \nu_i \quad \forall i \in S$. The *time-homogenous* property means that the conditional probability doesn't change with regard to time i.e. it doesn't change with respect to n . The *Markov property* states that the probabilities at time $n + 1$ depend only on the state at time n i.e.

$$\mathbb{P}(X_{n+1} = j | X_0 = x_0, X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_{n+1} = j | X_n = x_n) = p_{ij}$$

Using these two, we can compute the joint probability using Bayes' rule for conditional probability. For example, take $i, j \in S$,

$$\mathbb{P}(X_0 = i, X_1 = j) = \mathbb{P}(X_1 = j | X_0 = i) \mathbb{P}(X_0 = i) = p_{ij} \nu_i$$

Generally speaking, we can recursively apply this definition to get the following:

Corollary 1.1.2 (Factorization of joint distribution).

$$\mathbb{P}(X_0 = x_0, X_1 = x_1, \dots, X_n = x_n) = \nu_{x_0} p_{x_0 x_1} p_{x_1 x_2} \cdots p_{x_{n-1} x_n}$$

The probabilities defined on the sequence $\{X_n\}_{n=0}^{\infty}$ establishes the definition of a Markov chain.

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1.1.2 Multistep transitions

As an example, take $i, j, k \in S$ and consider the following:

$$\begin{aligned}\mathbb{P}(X_2 = j | X_0 = i) &= \sum_{k \in S} \mathbb{P}(X_2 = j | X_1 = k, X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_2 = j | X_1 = k) \mathbb{P}(X_1 = k | X_0 = i) \\ &= \sum_{k \in S} p_{ik} p_{kj}\end{aligned}$$

Remark 1.1.3. When $|S| < \infty$, this is just matrix-multiplication. ie. $\mathbb{P}(X_2 = j | X_0 = i) = (\mathbf{P}^2)_{i,j}$. This is also true for the countably-infinite case. In particular, if $\mathbf{A} = (a_{ij})_{i,j \in S}$ and $\mathbf{B} = (b_{ij})_{i,j \in S}$ are transition matrices in countably infinite state space S , then

$$(\mathbf{AB})_{i,j} = \sum_{k \in S} a_{ik} b_{kj}$$

and we can show $\mathbf{AB}$ is also a transition matrix on S which corresponds to the composition of 2-step transitions.

In general, by induction we have

- $\mathbb{P}(X_n = j | X_0 = i) = (\mathbf{P}^n)_{i,j}$
- $\mathbb{P}(X_n = j) = \sum_{i \in S} \nu_i (\mathbf{P}^n)_{i,j} = (\nu \cdot \mathbf{P}^n)_j$ ¹

The notation going forward will use $p_{ij}^{(n)} := (\mathbf{P}^n)_{i,j} = \mathbb{P}(X_n = j | X_0 = i)$

1.1.3 Chapman-Kolmogorov Equation

Theorem 1.1.4 (Chapman-Kolmogorov Equations).

$$p_{ij}^{(m+n)} = (\mathbf{P}^{m+n})_{i,j} = (\mathbf{P}^m \mathbf{P}^n)_{i,j} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)} \quad (1.1)$$

Proof. Using the Law of Total Probability, and Markov property, we get the following:

$$\begin{aligned}p_{ij}^{(m+n)} &= \mathbb{P}(X_{m+n} = j | X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{m+n} = j | X_m = k, X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{m+n} = j | X_m = k) \mathbb{P}(X_m = k | X_0 = i) \\ &= \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}\end{aligned}$$

□

In the proof of the Chapman-Kolmogorov equation, we fixed an indeterminate time m , and applied the Markov property for the proof. However, if we wanted to use a random indeterminate time, the standard Markov property is no longer sufficient. In such a case, we will give a loose idea of the "Strong Markov property".

¹This is a $1 \times S$ vector, and you take the j th element where ν is just a row vector that is also $1 \times S$

Theorem 1.1.5 (Strong Markov property). *Let $\{X_n\}_{n=0}^\infty$ be Markov(λ, P) and let τ be a stopping time of $\{X_n\}_{n=0}^\infty$. Then, conditional on $\tau < \infty$ and $X_\tau = i$, $\{X_{n+\tau}\}_{n \geq 0}$ is Markov(δ_i, P) and independent of $X_0, X_1, \dots, X_\tau$*

For instance, the first passage of time. ie $\tau = \inf\{t : X_t = i\}$ is a stopping time, and one can show the Strong Markov property will hold at all stopping times ². In plain English, starting from X_0 would be some random length, and at time τ where $X_\tau = i$ would be a fresh new Markov chain starting from state i . However, this is non-trivial to show. In general, the Strong Markov property holds for discrete-time Markov chains in discrete spaces.

1.2 Recurrence and Transience

Generally, we care about long-run behavior of Markov chains. We want to know the limiting behavior, and convergence/divergence. Is there a state visited infinitely often? or an absorbing state?

For a state $i \in S$, let $N(i) := \sum_{t=1}^{\infty} \mathbf{1}_{\{X_t=i\}}$ i.e. the total # of visits to state i in the Markov chain trajectory. Note that $N(i)$ is also a random variable depending on infinite MC trajectory. We also denote $f_{ij} := \mathbb{P}(N(j) \geq 1 | X_0 = i)$ to be the return probability from i to j . It can be thought of as the probability of visiting state j if we start at state i .

Definition 1.2.1 (Recurrent and transient states). A state $i \in S$ is recurrent if $f_{ii} = 1$ and transient if $f_{ii} < 1$.

Intuitively speaking, recurrence means that if we start at state i , eventually it will go back to state i a.s. (reoccurring). Transient means if we start state i , there is some positive probability you never go back (don't reoccur). Additionally, we denote $i \rightarrow j$ to be the fact that $f_{ij} > 0$ i.e. $\exists$ path with positive probability from state i to state j . The following characterizes the probability of hitting a state k times.

Theorem 1.2.2. *For a state $i \in S$,*

$$\mathbb{P}_i(N(i) \geq k) = (f_{ii})^k \quad (1.2)$$

³ and similarly,

$$\mathbb{P}_i(N(j) \geq k) = f_{ij}(f_{jj})^k \quad (1.3)$$

- if $f_{ii} = 1$, it is recurrent and $N(i) = \infty$ a.s.
- if $f_{ii} < 1$, $N(i) \sim \text{Geom}(f_{ii})$

Proof. We will use an induction argument to prove the first equation, the second can be proven using a similar argument.

If $k = 1$, then $\mathbb{P}_i(N(i) \geq 1) = f_{ii}$ by definition

Assume that $\mathbb{P}_i(N(i) \geq k) = f_{ii}^k$ for certain $k \geq 0$ [IH]

For the case of $k + 1$:

$$\mathbb{P}_i(N(i) \geq k + 1) = \underbrace{\mathbb{P}_i(N(i) \geq k)}_{=f_{ii}^k \text{ by IH}} \cdot \underbrace{\mathbb{P}_i(N(i) \geq k + 1 | N(i) \geq k)}_{\text{WTS}=f_{ii}}$$

²Instances of stopping times include hitting time of state i or the k th hitting time of state i

³ $\mathbb{P}_i(\dots)$ is shorthand notation to denote $\mathbb{P}(\dots | X_0 = i)$

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Define $\tau_i^{(k)} := k\text{th hitting time of } i \text{ } (\tau_i^{(k)} = \inf\{t > \tau_i^{(k-1)} : X_t = i\})$. We know that the event $\{N(i) \geq k\} = \{\tau_i^{(k)} < \infty\}$. Naturally, we need to compute

$$\mathbb{P}(\tau_i^{(k+1)} < \infty | \tau_i^{(k)} < \infty)$$

We have that:

$$\begin{array}{ccccccccc} X_0, X_1, \dots, X_{\tau_i^{(1)}}, \dots, X_{\tau_i^{(2)}}, \dots, X_{\tau_i^{(k)}}, \dots, X_{\tau_i^{(k+1)}}? \\ \parallel & & \parallel & & \parallel & & \parallel & & \parallel \\ i & & i & & i & & i & & i \end{array}$$

By the Strong Markov property, we can condition starting from time $t = \tau_i^{(k)}$, and we know that $\{X_t\}_{t=\tau_i^{(k)}}$ is a new MC starting from state i and that $\{X_t\}_{t=\tau_i^{(k)}} \stackrel{d}{=} \{X_t\}_{t \geq 0}$.

So $\mathbb{P}(\tau_i^{(k+1)} < \infty | X_0, X_1, \dots, X_{\tau_i^{(k)}}) = \mathbb{P}_i(\tau_i^{(1)} < \infty) = f_{ii}$. Substituting this back completes the proof. $\square$

For a Markov chain starting at i , if $f_{ii} < 1 \implies \mathbb{P}(N(i) = k) = f_{ii}^k (1 - f_{ii})$ ⁴ and we know $\mathbb{E}_i[N(i)] = \frac{f_{ii}}{1-f_{ii}}$ which is a geometric distribution. Furthermore, we can also compute

$$\begin{aligned} \mathbb{E}_i[N(j)] &= \mathbb{P}_i(N(j) \geq 0)(1 + \mathbb{E}_j[N(j)]) \\ &= f_{ij} \left(1 + \frac{f_{jj}}{(1 - f_{jj})} \right) \\ &= \frac{f_{ij}}{(1 - f_{jj})} \quad \text{when } f_{jj} < 1 \end{aligned}$$

There are two "corner cases" that still need to be considered:

- if $f_{ij} = 0$, then $\mathbb{E}_i[N(j)] = 0$
- if $f_{ij} > 0, f_{jj} = 1$, then $\mathbb{E}_i[N(j)] = \infty$

1.3 Recurrent State Theorem

A natural question that arises now is if we can say something about recurrence/transience directly from $\mathbf{P}$ itself. The following result characterizes the recurrence of a state i :

Theorem 1.3.1 (Recurrent State Theorem). *A state $i \in S$ is recurrent iff $\mathbb{P}_i(N(i) = \infty) = 1$ iff $\sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty$ and similarly i is transient iff $\mathbb{P}_i(N(i) = \infty) = 0$ iff $\sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty$*

Proof. Let $E_n = \{X_n = i\}$, ie. we visit state i at time n . If $\sum_{n=1}^{\infty} \mathbb{P}(E_n) = \sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty$, then by the Borel-Cantelli lemma, $\{E_n\}_{n=1}^{\infty}$ happens only finitely often *w.p.1*. Thus implying transience and proving one direction. From the proof of Borel-Cantelli, its clear that the converse isn't guaranteed, there exists real-values random variables with infinite expectation. However, for Markov Chains, $N(i) = \sum_{i=1}^n \mathbf{1}_{\{X_n=i\}}$ is either infinite *w.p.1* or finite, and geometrically distributed. Using this

⁴ $(1 - f_{ii})$ is the probability that it doesn't return to state i starting from i

observation we can prove the other direction of the recurrent state theorem. By Fubini-Tonelli we get the following:

$$\begin{aligned}\sum_{i=1}^n p_{ii}^{(n)} &= \mathbb{E}_i \left[\sum_{i=1}^n \mathbf{1}_{\{X_n=i\}} \right] \\ &= \mathbb{E}_i[N(i)]\end{aligned}$$

if i is transient, then $f_{ii} < 1$, and recall that $\mathbb{E}_i[N(i)] = \frac{f_{ii}}{1-f_{ii}} < \infty$ □

1.4 f-expansion

The next question that arises is how do we compute f_{ij} if we can compute f_{ij} systemetically then we know which states are recurrent/transient by definition.

Example 1.4.1. From the above diagram we see that: $f_{11} = 1$, $f_{1,j} = 0$ for $j \in \{2, 3, 4\}$ $f_{22} = \frac{1}{3}$, and $f_{21} = \frac{1}{3} + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \dots = \frac{1/3}{1-1/3} = 1/2$. Similarly, $f_{23} = f_{24}$ (if you get to 3, you get to 4), and $f_{23} = f_{24} = f_{21} = \frac{1}{2}$ (branches behave in a symmetric way).

We don't like the infinite sum structure, so in general, we use f-expansion i.e. solving our f_{ij} using linear systems.

Theorem 1.4.2 (f-expansion). *For states $i, j \in S$*

$$f_{ij} = p_{ij} + \sum_{k \in S, k \neq j} p_{ik} f_{kj}$$

Proof.

$$\begin{aligned}f_{ij} &= \mathbb{P}_i(\text{visit } j) \\ &= \sum_{k \in S} \mathbb{P}_i(X_1 = k) \cdot \mathbb{P}_k(\text{visit } j \text{ starting from time } t = 0) \\ &= p_{ij}(1) + \sum_{k \in S, k \neq j} p_{ik} f_{kj}\end{aligned}$$

□

Intuitively, we can either go directly from i to j with probability p_{ij} , or from i we can go to another state k with probability p_{ik} , then go to j from k with probability p_{kj} . If $|S| < \infty$, f-expansion gives $|S|^2$ equations for $|S|^2$ variables, and we can solve out these variables. We turn to an interesting problem involving Markov Chains.

Example 1.4.3 (Gambler's Ruin). Suppose a gambler plays coin-tossing games.

- Initially, $a \in \mathbb{N}^+$ units of money
- For each round, bet 1-unit with winning probability $p \in (0, 1)$
- Stop when you lose all your money or your money hits some threshold $c > a$ (Wise gambler?)

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In markov chain language, let X_n be the amount of money in n th round and $X_n \in \{1, 2, \dots, c-1\}$

$$X_0 = a \quad X_{n+1} = X_n + \epsilon_{n+1} \quad \text{where} \quad \epsilon_{n+1} = \begin{cases} +1 & \text{w.p } p \\ -1 & \text{w.p } 1-p \end{cases}$$

and if $X_n \in \{0, c\}$, then the game ends and $X_{n+1} = X_n$.

We are interested in the ruin probability i.e.

$$\mathbb{P}_a(\text{visit } 0) = f_{a0}$$

We have $f_{00} = 1, f_{c0} = 0$. By f-expansion, we have:

$$f_{a0} = pf_{(a+1)0} + (1-p)f_{(a-1)0} \quad (\text{when } a \geq 1)$$

Case 1: $p = \frac{1}{2}$

Setting $p = 1/2$, solving the f-expansion equations gives:

$$f_{a0} = \frac{1}{2} (f_{(a+1)0} + f_{(a-1)0})$$

So, $(f_{a0})_{a=0}^c$ is an arithmetic sequence.

$$f_{c0} - f_{(c-1)0} = f_{(a+1)0} - f_{a0} = f_{a0} - f_{(a-1)0} = \dots = f_{10} - f_{00} = -\frac{1}{c}$$

We know that $f_{00} = 1, f_{10} = 0$, and therefore, $f_{a0} = 1 - \frac{a}{c}$. Similarly, $f_{ac} = \frac{a}{c}$

Remark 1.4.4. We cannot make money from it in expectation.

$$\mathbb{E}[X_{final}] = c \cdot \frac{a}{c} + 0 \left(1 - \frac{a}{c}\right) = a$$

In general, this is not a coincidence c.f. martingales.

Case 2: $p \neq \frac{1}{2}$ Recall, in case 1:

$$\begin{aligned} f_{a0} &= pf_{(a+1)0} + (1-p)f_{(a-1)0} \\ pf_{(a+1)0} - pf_{a0} + (1-p)f_{(a-1)0} - (1-p)f_{a0} &= 0 \\ \implies f_{(a+1)0} - f_{a0} &= \frac{(1-p)}{p} (f_{a0} - f_{(a-1)0}) \end{aligned}$$

Notice that the increment forms a geometric sequence, so we have

$$f_{(a+1)0} - f_{a0} = \left(\frac{(1-p)}{p}\right)^a (f_{10} - f_{00})$$

Summing over a ,

$$\begin{aligned} -1 &= f_{10} - f_{00} = \sum_{a=0}^{c-1} (f_{(a+1)0} - f_{a0}) \\ &= (f_{10} - f_{00}) \sum_{a=0}^{c-1} \left(\frac{(1-p)}{p}\right)^a \end{aligned}$$

Following some calculation,

$$f_{a0} = \frac{\left(\frac{(1-p)}{p}\right)^c - \left(\frac{(1-p)}{p}\right)^a}{\left(\frac{(1-p)}{p}\right)^c - 1}$$

Remark 1.4.5. When $p \neq \frac{1}{2}$ the ruin probability becomes exponentially small when $p > \frac{1}{2}$ and exponentially close to 1 when $p < \frac{1}{2}$

1.5 Equivalent characterizations of recurrence/transience

Recall, that $i \rightarrow j$ means j is reachable from i , $f_{ij} > 0$. $i \leftrightarrow j$ means $i \rightarrow j$ and $j \rightarrow i$.

Definition 1.5.1 (Irreducibility of Markov Chains). We call a Markov Chain irreducible when $i \leftrightarrow j$ for $i, j \in S$. (Implicitly, an MC not satisfying this can be "reduced" into smaller ones).

The following lemma will be used to help us prove an important theorem for irreducible markov chains, and can also be helpful for checking the condition $\sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty$ in the reccurent state theorem from earlier.

Lemma 1.5.2 (Sum Lemma). *If i, j, k, l in S (allow repeating). Suppose $i \rightarrow k$ and $l \rightarrow j$. If $\sum_{n=1}^{\infty} p_{kl}^{(n)} = \infty$ then $\sum_{n=1}^{\infty} p_{ij}^{(n)} = \infty$*

Proof. Since $i \rightarrow k, \exists m > 0$ s.t. $p_{ik}^{(m)} > 0$, and $l \rightarrow j, \exists r > 0$ s.t. $p_{lj}^{(r)} > 0$. Then by Chapman-Kolmogorov equations, we have that

$$p_{ij}^{((n+m+r))} \geq p_{ik}^{((m))} p_{kl}^{((n))} p_{lj}^{((r))}$$

Summing this together,

$$\sum_{n=1}^{\infty} p_{ij}^{(n)} \geq p_{ik}^{(m)} p_{lj}^{(r)} \sum_{n=1}^{\infty} p_{kl}^{(n)} = \infty$$

□

Theorem 1.5.3 (Recurrence Equivalence). *If $i \leftrightarrow j$, then "i is recurrent" $\iff$ "j is recurrent"*

Proof. Applying Sum lemma with $j = i, l = k, i \rightarrow k, l \rightarrow j \iff i \leftrightarrow l$. We know that l is recurrent $\iff \sum_{n=1}^{\infty} p_{ll}^{(n)} = \infty$. The Sum lemma implies $\sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty$ □

The following is a useful theorem which is a corollary of the above theorem.

Corollary 1.5.4 (Case Theorem of Markov Chains). *If a Markov Chain is irreducible, then one of the following is true:*

- all states are recurrent ("recurrent Markov Chain")
- all states are transient ("transient Markov Chain")

The next question that arises is "What do we know about $\sum_{n=1}^{\infty} p_{ij}^{(n)}$?" (for $i \neq j$)

Theorem 1.5.5. *if Markov Chain is recurrent, then $\sum_{n=1}^{\infty} p_{ij}^{(n)} = \infty$. If a Markov Chain is transient, then $\sum_{n=1}^{\infty} p_{ij}^{(n)} < \infty$. (This is iff, any i, j pair is equivalent to Markov Chain recurrence/transience)*

Proof.

$$\sum_{n=1}^{\infty} p_{ij}^{(n)} = \mathbb{E}_i[N(j)]$$

- Recurrent, $\exists m$ s.t. $p_{ij}^{(m)} > 0$ and by irreducibility $\mathbb{E}_i[N(j)] \geq p_{ij}^{(m)} \mathbb{E}_j[N(j)] = \infty$
- Transient, $\mathbb{E}_i[N(j)] = f_{ij} \mathbb{E}_j[N(j) + 1] = \frac{f_{ij}}{1 - f_{jj}} < \infty$

□

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Example 1.5.6. When $|S| < \infty$, and irreducible, then the Markov Chain is recurrent.

Proof. For any fixed state $i \in S$. Using Fubini-Tonelli,

$$\sum_{j \in S} \left(\sum_{n=1}^{\infty} p_{ij}^{(n)} \right) = \sum_{n=1}^{\infty} \sum_{j \in S} p_{ij}^{(n)} = \sum_{n=1}^{\infty} 1 = \infty$$

and by irreducibility, this implies that the Markov Chain is recurrent. $\square$

Another useful characterization for recurrence/transience is the f-lemma. To prove the f-lemma, we will need to use a technical tool: "hit lemma". Define the event

$$H_{ij} = \{\text{Markov Chain hits } i \text{ before returning to } j\}$$

Lemma 1.5.7 (hit-lemma). *If $j \rightarrow i$, then $\mathbb{P}_i(H_{ij}) > 0$*

Proof. $j \rightarrow i$, so $\exists x_0, x_1, \dots, x_\ell, \dots, x_{m-1}, x_m$ where $x_0 = j, x_\ell = j, x_m = i$. We have

$$p_{x_0, x_1}, p_{x_1, x_2}, \dots, p_{x_{m-1}, x_m} > 0$$

This implies that

$$p_{x_\ell, x_{\ell+1}}, \dots, p_{x_{m-1}, x_m} > 0$$

Let ℓ be the last time j appears in the sequence.

$$\mathbb{P}_j(H_{ij}) \geq p_{x_\ell, x_{\ell+1}}, \dots, p_{x_{m-1}, x_m} > 0$$

$\square$

Lemma 1.5.8 (f-lemma). *if $j \rightarrow i, f_{jj} = 1 \Rightarrow f_{ij} = 1$ a direct corollary of this is as follows. For recurrent Markov Chains, $f_{ij} = 1 \forall i, j \in S$*

Proof.

$$0 = \mathbb{P}_j(\text{never return to } j) \geq \underbrace{\mathbb{P}_j(H_{ij})}_{>0} \underbrace{\mathbb{P}_i(\text{never visit } j)}_{\text{must be } 0}$$

So, $f_{ij} = 1$ which proves f-lemma. $\square$

Before we get to the equivalence condition for recurrent chains. We need one more tool: "Infinite returns lemma".

Lemma 1.5.9 (Infinite returns lemma). *For an irreducible Markov Chain, if it is recurrent then $\forall i, j \in S \mathbb{P}_i(N(j) = \infty) = 1$. But if it is transient, then $\forall i, j \in S \mathbb{P}_i(N(j) = \infty) = 0$.*

Proof. For recurrent case, by f-lemma $f_{ij} = 1$, starting from $j \mathbb{P}(N(j) = \infty) = 1$. For the transient case, $\mathbb{E}_i[N(j)] = \frac{f_{ij}}{1-f_{jj}} < \infty$ $\square$

Putting all these results together, we have the following equivalent conditions for recurrent chains.

Theorem 1.5.10 (Recurrence Equivalences Theorem). *If a chain is irreducible, then we have the following equivalences.*

$$(i) \text{ Recurrent } \iff \forall \text{ or } \exists i, j \in S \text{ such that } \sum_{n=1}^{\infty} p_{ij}^{(n)} = \infty$$

(ii) *Recurrent* $\iff \forall$ or $\exists i, j \in S$ such that $\mathbb{P}_i(N(j) = \infty) = 1$

(iii) *Recurrent* $\iff \forall$ or $\exists i \in S$ such that $f_{ii} = 1$

(iv) *Recurrent* $\iff \forall i, j \in S, f_{ij} = 1$

Proof. (i) follows from sum lemma. (ii) follows from Infinite returns lemma. (iii) follows from Recurrent state theorem. (iv) follows from the f-lemma. $\square$

Some of you may be wondering if we are able to conclude recurrence from the condition that $\exists i, j \in S, f_{ij} = 1$. However, without additional assumptions, this condition alone isn't strong enough. We can construct a counterexample, where the Markov Chain is irreducible such that $\exists i, j \in S, f_{ij} = 1$ but it is transient.

Example 1.5.11 (Biased SRW). Let $X_0 = 0$, and consider the following process:

$$X_{n+1} = X_n + \epsilon_{n+1} \quad \text{where} \quad \epsilon_{n+1} = \begin{cases} +1 & \text{w.p. } 2/3 \\ -1 & \text{w.p. } 1/3 \end{cases}$$

First, this is irreducible. Second, since we know that

$$p_{0,0} = \begin{cases} 0 & (\text{n is odd}) \\ \binom{n}{n/2} \left(\frac{1}{3}\right)^{n/2} \left(\frac{2}{3}\right)^{n/2} & (\text{n is even}) \end{cases}$$

By Stirling's approximation, we know that this is $\approx \frac{c}{\sqrt{n}} \left(\frac{8}{9}\right)^{n/2}$. So we know that $\sum_{n=1}^{\infty} p_{00}^{(n)} < \infty$ which implies transience.

Claim: $f_{01} = 1$

Proof. by SLLN, $\frac{X_n}{n} \xrightarrow{a.s.} \mathbb{E}[\epsilon_i] = \frac{1}{3} > 0$. Since

$$X_n = \sum_{i=1}^{n-1} \epsilon_i$$

We know that $X_n \rightarrow \infty$ w.p.1. On this probability 1 event, $(X_n)_{n \geq 1}$ must visit 1, so $f_{01} = 1$ $\square$

On the other hand, if we assume that

$$f_{ij} = f_{ji} = 1 \quad \text{for some } i, j \in S \implies f_{ii} = 1$$

The intersection of both events is 1, and therefore we can conclude recurrence.

Since transience is the complement of recurrence, the transience equivalence theorem is as follows:

Theorem 1.5.12 (Transience Equivalences Theorem). *If a chain is irreducible, then we have the following equivalences.*

(i) *Transience* $\iff \forall$ or $\exists i, j \in S$ such that $\sum_{n=1}^{\infty} p_{ij}^{(n)} < \infty$

(ii) *Transience* $\iff \forall$ or $\exists i, j \in S$ such that $\mathbb{P}_i(N(j) < \infty) = 1$

(iii) *Transience* $\iff \forall$ or $\exists i \in S$ such that $f_{ii} < 1$

(iv) *Transience* $\iff \exists i, j \in S, f_{ij} < 1$

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Proof. See Recurrence Equivalences Theorem. □

For a reducible Markov Chain consider the following example. For finite state space, structurally like gambler's ruin. We can rearrange the matrix into the following block matrix.

$$P = \left(\begin{array}{cccc|c} p_1 & 0 & \dots & 0 & \\ 0 & p_2 & \dots & 0 & \mathbf{O} \\ \vdots & \vdots & \ddots & \vdots & \\ 0 & 0 & \dots & p_k & \\ \hline & \mathbf{S} & & & \mathbf{Q} \end{array} \right) \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} \text{recurrent states} \\ \\ \\ \end{array} \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} \\ \\ \\ \text{transient states} \end{array}$$

It is impossible to go from recurrent to transient state (f-lemma). If $f_{ii} = 1, f_{ij} > 0$, then j is recurrent. Which recurrent state do you go to?

1.6 Stationary distributions

The next question we want to answer is the following:

$$\lim_{n \rightarrow \infty} \mathbb{P}_i(X_n = j) \text{ or } \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \mathbb{P}(X_i = j)$$

Suppose convergence occurs, $p_{ij}^{(n)} \rightarrow q_j \quad \forall i, j \in S$. Then by Chapman-Kolmogorov, we know that $p_{ij}^{(n+1)} = \sum_{k \in S} p_{ik}^{(n)} p_{kj}$. Taking limits of both sides we get

$$q_j = \sum_{k \in S} q_k p_{kj} \quad \forall j \in S$$

i.e. $q = qP$

Definition 1.6.1 (Stationary Distribution). We say π is a stationary distribution of Markov Chain P when

- π is a probability distribution
- $\pi = \pi P$ thinking π as a row vector: $(\pi_j = \sum_{i \in S} \pi_i p_{ij} \quad \forall j \in S)$

Relaxing the first condition, we have the following definition:

Definition 1.6.2 (Stationary Measure). We say π is a stationary measure for P if

- $\pi(j) \geq 0 \quad \forall i, \pi \neq 0, \sum_{i \in S} \pi(i)$ can be infinite (but not necessarily a stationary distribution)
- $\pi = \pi P$

Example 1.6.3 (Frog Walk on 20 states). The setup can be seen in [Ros06] Example 1.1.

$$\pi = \left[\frac{1}{20}, \dots, \frac{1}{20} \right] \quad (\text{uniform distribution})$$

Easy to verify that for any state $i \in S$

$$\pi_i = \frac{1}{20} = \sum_{j \in S} \pi_j p_{ji} = \frac{1}{20} \cdot \frac{1}{3} \cdot 3 = \frac{1}{20}$$

Example 1.6.4 (Doubly Stochastic matrices). A matrix is doubly stochastic if

$$\forall i \in S, \quad \sum_{j \in S} p_{ij}, \quad \sum_{j \in S} p_{ji} = 1$$

If P is doubly stochastic ($|S| < \infty$) then uniform distribution is stationary. This extends to stationary measures when $|S| = \infty$

Proof. $\frac{1}{|S|} = \pi_i \implies \sum_{j \in S} \pi_j p_{ji} = \sum_{j \in S} \frac{1}{|S|} p_{ji} = \frac{1}{|S|} \sum_{j \in S} p_{ji} = \frac{1}{|S|}$ □

Example 1.6.5 (Simple random walk in 1-D).

$$\pi(i) = 1 \quad \forall i \in S$$

is a stationary measure

To find a stationary distribution/measure the following can be helpful.

Definition 1.6.6 (Detailed-Balance Condition/Reversibility). A Markov Chain satisfies the detailed balance condition/is reversible with respect to a probability distribution if

$$\pi_i p_{ij} = \pi_j p_{ji} \quad \forall i, j \in S$$

Proposition 1.6.7. If P is reversible w.r.t $\pi \implies \pi$ is a stationary distribution/measure for P

Proof. If P is reversible then $\pi_i p_{ij} = \pi_j p_{ji}$, so for $j \in S$, $\sum_{i \in S} \pi_i p_{ij} = \sum_{i \in S} \pi_j p_{ji} = \pi_j \sum_{i \in S} p_{ji} = \pi_j$. Thus, holds by definition. □

Example 1.6.8 (Biased RW).

$$X_{n+1} = X_n + \epsilon_{n+1} \quad \text{where} \quad \epsilon_{n+1} = \begin{cases} +1 & \text{w.p } p \\ -1 & \text{w.p } 1-p \end{cases}$$

There is not a stationary distribution, but we can construct a stationary measure. Let $\pi(i) = \left(\frac{p}{1-p}\right)^i$.

$$\pi_i p_{i(i+1)} = \left(\frac{p}{1-p}\right)^i \cdot p = \frac{p^{i+1}}{(1-p)^i}$$

$$\pi_{i+1} p_{(i+1)i} = \left(\frac{p}{1-p}\right)^{i+1} (1-p) = \frac{p^{i+1}}{(1-p)^i}$$

Therefore, the detailed balance condition holds.

Example 1.6.9 (Ehrenfest urn). An example and simulation of Ehrenfest's urn can be seen here.

X_n = number of balls in box 1 in n th round

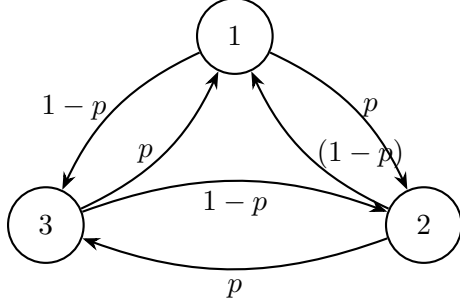
Conceptually, balls go back and forth. Guess: stationary/limiting distribution uniform into each other. $\pi(i) = \binom{d}{i} 2^{-d}$ for $i = 0, \dots, d$. Intuitively, each ball iid thrown into a box.

$$\begin{aligned} \pi_i p_{i(i+1)} &= \binom{d}{i} 2^{-d} \frac{d-i}{d} \\ &= \frac{d!}{i!(d-i)!} \cdot 2^{-d} \cdot \frac{d-i}{d} = \frac{(d-1)! 2^{-d}}{i!(d-1-i)!} \end{aligned}$$

$$\begin{aligned}
\pi_{i+1}p_{(i+1)i} &= \frac{d!}{(i+1)!(d-1-i)!} \cdot 2^{-d} \cdot \frac{i+1}{d} \\
&= \frac{(d-1)!2^{-d}}{i!(d-1-i)!}
\end{aligned}$$

Therefore, detailed balance condition is satisfied.

Example 1.6.10 (MC P with stationary distribution π but detailed balance condition is false).



$\pi_i = \frac{1}{3} \quad \forall i \in S$ but detailed balance condition is false when $p \neq \frac{1}{2}$. Interestingly, if $p = 1$ then we have a degenerate Markov Chain and this has a unique stationary distribution. From a linear algebra perspective, the detailed balance condition eigenvalue of transition kernel are real, construct directly to stationary distribution without rotating.

For the existence/non-existence of a stationary distribution/measure, the following are helpful to check.

Corollary 1.6.11. *If $|S| < \infty$, then stationary distribution always exists*

Proof. The proof involves using Brouwer's fixed point theorem, if you have a convex and compact domain then you always have a fixed point. For infinite spaces they aren't compact, and note that this doesn't guarantee a unique stationary distribution (ex. reducible MC). $\square$

Theorem 1.6.12 (Vanishing probabilities). *if $\forall i, j \in S, \lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0$ then $\nexists$ stationary distribution for Markov Chain.*

Proof. Suppose a stationary distribution exists call it π

$$\pi = \pi P = \pi P^2 = \dots = \pi P^n = \dots$$

We want to show that $\pi_j = \sum_{i \in S} \pi_i p_{ij}^{(n)} \rightarrow \sum_{i \in S} \pi_i \lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0$ ie. we need to justify swapping the summation and limit. Recall Weierstrass M-test, for any infinitely dimensional matrix $\{X_{n,k}\}_{n,k \in \mathbb{N}}$. Suppose $\forall k, \lim_{n \rightarrow \infty} X_{n,k}$ exists and $\sum_{k=1}^{\infty} \sup_{n \geq 1} |X_{n,k}| < \infty$ then $\lim_{n \rightarrow \infty} \sum_{k=1}^{\infty} X_{n,k} = \sum_{k=1}^{\infty} \lim_{n \rightarrow \infty} X_{n,k}$. We know that $\sum_{i \in S} \sup_{n \geq 1} |\pi_i p_{ij}^{(n)}| \leq \sum_{i \in S} \pi_i = 1$. Since the LHS = 1, and the RHS = 0, for finite state space $|S|$, $\sum_{k \in S} p_{ij}^{(n)} = 1$, so $p_{ij}^{(n)}$ cannot go to 0. $\square$

The $\forall i, j$ condition isn't very nice to work with, we want to do the same for existence.

Lemma 1.6.13 (Vanishing Lemma). *If $k \rightarrow i, j \rightarrow l \lim_{n \rightarrow \infty} p_{k,l}^{(n)} = 0$ then $\lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0$*

Proof. Fix m, r , and suppose that $n \geq m + r$, we know that

$$p_{kl}^{(n)} \geq p_{ki}^{(m)} p_{ij}^{(n-m-r)} p_{jl}^{(r)}$$

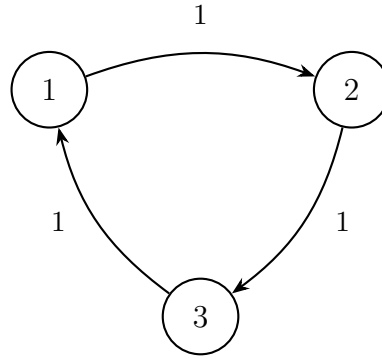
We know that the first term on the left hand side is positive and fixed, and the last term is also positive and fixed. By assumption we know that the left hand side goes to 0. Thus, as n goes to infinity, $p_{ij}^{(n-m-r)} \rightarrow 0$. $\square$

Corollary 1.6.14. *For irreducible MC, if $\exists k, l \in S$ such that $\lim_{n \rightarrow \infty} p_{kl}^{(n)} = 0$ then $\nexists$ stationary distribution.*

Corollary 1.6.15 (Vanishing together). *For irreducible MC one of the following happens*

1. $\forall i, j \in S \quad \lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0$
2. $\forall i, j \in S \quad \lim_{n \rightarrow \infty} p_{ij}^{(n)} \neq 0$ (doesn't mean it converges to something)

Note: in the latter case, limit may not converge exist. Consider the following case



Example 1.6.16 (3 State Degenerate MC).
Stationary exists and unique $\lim_{n \rightarrow \infty} p_{ij}^{(n)}$?

$$p_{11}^{(n)} = \begin{cases} 0 & \text{when } n/3 \notin \mathbb{Z} \\ 1 & \text{when } n/3 \in \mathbb{Z} \end{cases}$$

You will realize it never converges, but the area will converge. $\frac{1}{n} \sum_{t=1}^n p_{11}^{(t)} \rightarrow \frac{1}{3}$

To avoid the pathology above, we define the following:

1.6.1 Periodicity

Definition 1.6.17 (Period of a state). The period of a state $i \in S$ is $\gcd$ of $\{n \geq 1 : P_{ii}^{(n)} > 0\}$

Example 1.6.18. for the cyclic chain above, period of any state is 3

Example 1.6.19. if $p_{ii} > 0$, then the period of i is 1

Definition 1.6.20 (Aperiodic). A state is aperiodic iff period is equal to 1

Lemma 1.6.21 (Equal period lemma). *If $i \leftrightarrow j$ then i, j has the same period. Furthermore, in an irreducible chain all states have the same period.*

Proof. $p_{ij}^{(r)} > 0, p_{ji}^{(s)} > 0$

$$p_{ij}^{(r+n+s)} \geq p_{ij}^{(r)} p_{jj}^{(n)} p_{ji}^{(s)} > 0$$

$$p_{ii}^{(r+s)} \geq p_{ij}^{(r)} p_{ji}^{(s)} > 0 \quad \text{is also a path}$$

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Let t_i, t_j be period of i, j respectively

$$\frac{r+n+s}{t_i} \in \mathbb{Z}, \quad \frac{r+s}{t_i} \in \mathbb{Z}$$

So $\frac{n}{t_i} \in \mathbb{Z} \quad \forall n$ st. $p_{jj}^{(n)} > 0$. Thus, $t_i | t_j$, and by symmetry $t_j | t_i$ so $t_i = t_j$ □

Lemma 1.6.22 (Markov Forgetting Lemma). *Suppose P is irreducible and aperiodic with stationary distribution π . Then*

$$\forall i, j, k \in S \quad \lim_{n \rightarrow \infty} |p_{ik}^{(n)} - p_{jk}^{(n)}| = 0.$$

The proof proceeds in four steps: (0) two auxiliary facts about hitting positivity, (1) construction of an independent joint chain and proof that it is irreducible/aperiodic, (2) recurrence of the joint chain, giving a finite *coupling time*, and (3)–(4) the coupling argument itself.

Lemma 1.6.23 (Auxiliary Fact 1). *If state i is aperiodic and $f_{ii} > 0$, then $\exists n_0(i) \in \mathbb{N}_+$ such that*

$$p_{ii}^{(n)} > 0 \quad \forall n \geq n_0(i).$$

Proof. Let $A = \{n : p_{ii}^{(n)} > 0\}$. By the Chapman–Kolmogorov equations, if $m, n \in A$ then $m+n \in A$, since

$$p_{ii}^{(m+n)} \geq p_{ii}^{(m)} p_{ii}^{(n)} > 0.$$

So A is closed under addition, and by aperiodicity $\gcd(A) = 1$. By Bézout’s identity, a set of positive integers closed under addition with $\gcd$ equal to 1 contains all sufficiently large integers. Hence $A \supseteq \{n_0(i), n_0(i) + 1, \dots\}$ for some $n_0(i)$. □

Corollary 1.6.24 (Auxiliary Fact 2). *Under the irreducible and aperiodic assumption, $\forall i, j \in S$ there exists $n_0(i, j) > 0$ such that*

$$p_{ij}^{(n)} > 0 \quad \forall n \geq n_0(i, j).$$

Proof. By irreducibility, choose $r > 0$ with $p_{ij}^{(r)} > 0$. By Fact 1 applied to j , $p_{jj}^{(n)} > 0$ for all $n \geq n_0(j)$. Then for such n ,

$$p_{ij}^{(r+n)} \geq p_{ij}^{(r)} p_{jj}^{(n)} > 0.$$

Set $n_0(i, j) := r + n_0(j)$. □

Proof of the Forgetting Lemma. Compare two independent copies $\{X_n^{(1)}\}_{n \geq 0}$, started at $i \in S$, and $\{X_n^{(2)}\}_{n \geq 0}$, started at $j \in S$. Construct a Markov chain $\{\bar{X}_n\}_{n \geq 0} = \{(X_n^{(1)}, X_n^{(2)})\}_{n \geq 0}$ on $\bar{S} := S \times S$, starting from $(i, j) \in \bar{S}$, where the two coordinates evolve independently:

$$\bar{P}_{(i_1, i_2), (i_3, i_4)} = P_{i_1, i_3} P_{i_2, i_4}.$$

For any $(i, j), (k, l) \in \bar{S}$ and $n \geq \max\{n_0(i, k), n_0(j, l)\}$ (using Fact 2),

$$\bar{P}_{(i, j), (k, l)}^{(n)} = p_{ik}^{(n)} p_{jl}^{(n)} > 0.$$

This shows every state of $\bar{S}$ is reachable from every other, i.e. $\bar{P}$ is irreducible; taking $(k, l) = (i, j)$ shows every state of $\bar{S}$ is aperiodic. Hence $\bar{P}$ is irreducible and aperiodic.

If π is a stationary distribution for P , then $\bar{\pi}_{(i,j)} := \pi_i \pi_j$ is stationary for $\bar{P}$:

$$\sum_{(i,j)} \pi_i \pi_j P_{i,k} P_{j,l} = \left(\sum_i \pi_i P_{i,k} \right) \left(\sum_j \pi_j P_{j,l} \right) = \pi_k \pi_l.$$

Hence $\bar{X}$ is positive recurrent. Fix any $i_0 \in S$ and define the coupling time:

$$\tau := \inf\{n \geq 0 : (X_n^{(1)}, X_n^{(2)}) = (i_0, i_0)\}.$$

By recurrence of $\bar{X}$, $\tau < \infty$ almost surely, starting from any $(i, j) \in \bar{S}$.

Fix $k \in S$ and decompose on the value of τ :

$$p_{ik}^{(n)} = \mathbb{P}_{(i,j)}(X_n^{(1)} = k) = \underbrace{\sum_{m=1}^n \mathbb{P}_{(i,j)}(X_n^{(1)} = k, \tau = m)}_{(*)} + \sum_{m=n+1}^{\infty} \mathbb{P}_{(i,j)}(X_n^{(1)} = k, \tau = m).$$

For $m \leq n$, on the event $\{\tau = m\}$ both copies occupy i_0 at time m , so by the Markov property

$$\mathbb{P}_{(i,j)}(X_n^{(1)} = k \mid \tau = m) = p_{i_0 k}^{(n-m)} = \mathbb{P}_{(i,j)}(X_n^{(2)} = k \mid \tau = m).$$

Thus $(*)$ is exactly the same quantity whether computed for $X_n^{(1)}$ or $X_n^{(2)}$ once the two chains have met. Therefore $(*)$ cancels in the difference $p_{ik}^{(n)} - p_{jk}^{(n)}$, and only the tail terms survive:

$$\begin{aligned} |p_{ik}^{(n)} - p_{jk}^{(n)}| &= \left| \sum_{m=n+1}^{\infty} \mathbb{P}_{(i,j)}(X_n^{(1)} = k, \tau = m) - \mathbb{P}_{(i,j)}(X_n^{(2)} = k, \tau = m) \right| \\ &\leq 2 \sum_{m=n+1}^{\infty} \mathbb{P}_{(i,j)}(\tau = m) = 2 \mathbb{P}_{(i,j)}(\tau > n). \end{aligned}$$

We know, $\tau < \infty$ almost surely, so $\mathbb{P}_{(i,j)}(\tau > n) \rightarrow 0$ as $n \rightarrow \infty$. Hence

$$|p_{ik}^{(n)} - p_{jk}^{(n)}| \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

which proves the lemma. $\square$

In general, coupling construction is used to prove speed of convergence (ex. card shuffling), coupling in some clever way so that they meet as early as possible. Before proving the Markov Chain Convergence Theorem, we need one more result that focuses on stationary distributions and recurrence.

Theorem 1.6.25 (Stationary Recurrence Theorem). *If a Markov chain is irreducible and had a stationary distribution, then it is recurrent.*

Proof. Suppose a chain is irreducible and transient, then $\sum_{n=1}^{\infty} p_{ij}^{(n)} < \infty$ for all $i, j \in S$. Hence, $\lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0$ for all $i, j \in S$. By the Vanishing together corollary, there cannot be a stationary distribution. Thus, if a chain is irreducible and has a stationary distribution, then it cannot be transient and must be recurrent. $\square$

1.7 Markov Chain Convergence

Theorem 1.7.1 (Markov Chain Convergence Theorem). *If a Markov Chain is irreducible, aperiodic, and has a stationary distribution π_j then*

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j \text{ for all } i, j \in S \text{ and } \lim_{n \rightarrow \infty} \mathbb{P}(X_n = j) = \pi_j$$

for any initial distribution ν_j

Proof.

$$\left| p_{ij}^{(n)} - \pi_j \right| = \left| p_{ij}^{(n)} - \sum_{k \in S} \pi_k p_{kj}^{(n)} \right| \leq \sum_{k \in S} \pi_k \left| p_{ij}^{(n)} - p_{kj}^{(n)} \right|$$

By the Markov Forgetting Lemma, we know that $\lim_{n \rightarrow \infty} \pi_k \left| p_{ij}^{(n)} - p_{kj}^{(n)} \right| = 0$ for all $k \in S$. We need to justify swapping the limit and the sum using Weirstrauss M-test.

$$\sum_{k \in S} \pi_k \underbrace{\sup_{n \geq 0} \left| p_{ij}^{(n)} - p_{kj}^{(n)} \right|}_{\leq 1} \leq \sum_{k \in S} \pi_k = 1 < \infty$$

So we have $\lim_{n \rightarrow \infty} \left| p_{ij}^{(n)} - \pi_j \right| = 0$. Similarly, if MC starts from $X_0 \sim \nu$, we can use a similar method to show $\mathbb{P}(X_n = j) \rightarrow \pi_j$ $\square$

We will now consider periodic chains. We have seen examples of periodic chains such as a s.r.w or Ehrenfest's urn. Consider the following example

Example 1.7.2. $S = \{1, 2\}$, $\nu_1 = 1$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. This example is irreducible and the period of this chain is 2, and the discrete uniform distribution $\pi_1 = \pi_2 = \frac{1}{2}$ is stationary. We know that:

$$\mathbb{P}(X_n = j) = \begin{cases} 1, & n \text{ is even} \\ 2, & n \text{ is odd} \end{cases}$$

Its clear that the transition probabilities don't converge to a stationary distribution. However, if we average the transition probabilities over different times (odd and even), then the average transition probabilities do converge to π . $\lim_{n \rightarrow \infty} \frac{1}{2} (p_{ij}^{(n)} + p_{ij}^{(n+1)}) = \pi_j = \frac{1}{2}$.

We will show that this type of convergence holds in general.

Theorem 1.7.3 (Periodic Convergence theorem). *Suppose a Markov chain is irreducible, with period $b \geq 2$, and has stationary distribution π_j . Then for all $i, j \in S$:*

$$\lim_{n \rightarrow \infty} \frac{1}{b} (p_{ij}^{(n)} + p_{ij}^{(n+1)} + \dots + p_{ij}^{(n+b-1)}) = \pi_j$$

Proof. (idea) This proof involves more number theory and combinatorics ideas. Take a "cyclic decomposition", by taking S disjoint subsets of the chain. Notice that

$$\pi(S_0) = \pi(S_1) = \dots = \pi(S_{b-1}) = \frac{1}{b}$$

. The key observation is that P^b is a MC on S_i for each i irreducible, aperiodic, and $\exists$ stationary. Then invoke previous theorem. $\square$

Corollary 1.7.4. *If P is irreducible, and has a stationary distribution π then $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=1}^n p_{ij}^{(m)} = \pi_j$*

Proof. Left as an exercise for the reader. Note that this also implies uniqueness of π as well. $\square$

1.8 Positive Recurrence

The next question is now related to the existence of stationary measures and distributions. When does a stationary measure become a stationary distribution?

Definition 1.8.1 (Positive Recurrence). A state $i \in S$ is positive recurrence if $\mathbb{E}_i[T_i] < \infty$, where T_i is the hitting time of i .

For the existence of stationary measures, we will show that recurrence is a sufficient condition.

Theorem 1.8.2. *If p is irreducible and recurrent, then fix $i_0 \in S$*

$$\forall j \in S, \mu_{i_0}(j) = \sum_{n=0}^{\infty} \mathbb{P}_{i_0}(X_n = j, T_{i_0} > n)$$

is positive and finite and μ_{i_0} is a stationary measure of MC.

Proof.

$$\mu_{i_0}(j) = \mathbb{E}_{i_0} [\# \text{ of visits to } j \text{ in } \{0, 1, \dots, T_{i_0} - 1\}]$$

$$(\mu_{i_0} P)(j) = \mathbb{E}_{i_0} [\# \text{ of visits to } j \text{ in } \{0, 1, \dots, T_{i_0}\}]$$

For $j \in S, j \neq i_0$, by stationarity

$$1 = \mu_{i_0}(i_0) = \sum_{j \in S} \mu_{i_0}(j) p_{ji_0}^{(n)}$$

By irreducibility,

$$\exists n \text{ s.t. } p_{ji_0}^{(n)} > 0$$

for this specific j . So, $\mu_{i_0}(j) \leq \frac{1}{p_{ji_0}^{(n)}} < \infty$ By hit lemma,

$$\mu_{i_0}(j) \geq \mathbb{P}_{i_0}(\text{visit } j \text{ before return to } i_0) > 0$$

$\square$

When does it become a stationary distribution?

$$\begin{aligned} \sum_{j \in S} \mu_i(j) &= \sum_{j \in S} \sum_{n=0}^{\infty} \mathbb{P}_i(X_n = j, T_i > n) \\ &= \sum_{n=0}^{\infty} \sum_{j \in S} \mathbb{P}_i(X_n = j, T_i > n) \\ &= \sum_{n=0}^{\infty} \mathbb{P}_i(T_i > n) \\ &= \mathbb{E}_i[T_i] \end{aligned}$$

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Immediately, when $\mathbb{E}_i[T_i] < \infty$ (positive recurrence)

$$\mu(j) = \frac{\mu_i(j)}{\mathbb{E}_i[T_i]} \forall j \in S$$

is a stationary distribution.

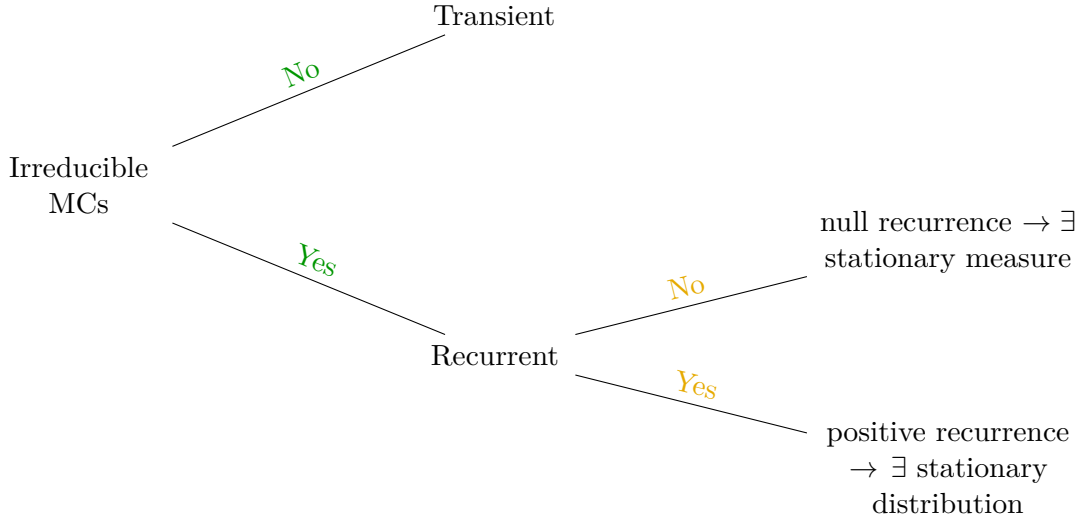
There are several structural properties of positive recurrence:

Lemma 1.8.3. *If $i \leftrightarrow j$, i is positive recurrent $\Rightarrow j$ is positive recurrent.*

Corollary 1.8.4. *P irreducible, $i \in S$ positive recurrent $\Rightarrow$ all states are positive recurrent (we call it a positive recurrent MC), and $\exists$ stationary distribution.*

Definition 1.8.5. $i \in S$ is null recurrent if $\mathbb{E}_i[T_i] = \infty$ we can show null recurrent chain does not have stationary distribution.

The following diagram can be quite useful to keep in mind from this section.



From Stationary distribution to $\mathbb{E}_i[T_i]$?

Theorem 1.8.6. *Suppose P is irreducible + recurrent*

$$N_n(i) = \sum_{t=1}^n \mathbf{1}_{X_t=i} \text{ then } \frac{N_n(i)}{n} \xrightarrow{\text{a.s.}} \frac{1}{\mathbb{E}_i[T_i]} \text{ (SLLN)}$$

Remark 1.8.7.

$$\left\{ \begin{array}{ll} \text{Null Recurrence} & \mathbb{E}_i[T_i] = \infty \iff \frac{N_n(i)}{n} \xrightarrow{\text{a.s.}} 0 \\ \text{Positive Recurrence} & \mathbb{E}_i[T_i] < \infty \iff \frac{N_n(i)}{n} \xrightarrow{\text{a.s.}} \frac{1}{\mathbb{E}_i[T_i]} \end{array} \right.$$

Chapter 2

Martingales

Chapter 3

Stochastic Calculus

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